



Idunn

Aras Janusonis (1), Ashley Jimenez (2), Curtis Whitt (3), Daniel Quezada Jimenez (4), Evert Johnson (5), Farah Job (6), Ivan Perez Jimenez (7), Julio Jimenez (8), Lourdes Johnson (9), Lukas Jaslonis (10), Ryan Johnston (11), Seungjun Jeong (12)

Computer Eng (1), Budget Officer (2), Systems Eng (3), Instrument Eng (4), Materials Eng (5), Science Communications Spc (6), Instrument Eng (7), Surface Operations Spc (8), Safety Officer (9), Mechanical Eng (10), Project Manager (11), Public Affairs (12)



Mission Goals and NASA Alignment

Idunn's mission is to send astronauts to the Moon, gain research on providing sustainable, nutritious, and delicious food in space, and return them safely.

- Our purpose is to maintain and improve health development in space to promote a commercialized and sustainable space environment.
- Idunn's goal is to conduct research that supports NASA's mission to understand how to sustain life on another celestial body for long periods of time.
- Our mission leveraged a collaboration with SpaceX, utilizing their rocket technology to meet the demands of our high-mass lunar payload.

Mission Design

Adequate Power

Idunn's energy system combines Regenerative Fuel Cells for high demands during lunar nights and MPRTGs for constant, reliable power. This system ensures seamless operation under all lunar conditions. With a total power 110 units aligning with NASA's sustainable exploration goals.

Successful Launch

To achieve a successful launch, our vehicle needed to generate sufficient lift to carry the full mission payload beyond Earth's atmosphere. Our team's total payload reached 237 mass units, just under the 240-unit limit, so we selected a rocket specifically aligned with our lift requirements to ensure mission success.

Science Return

Our team made having a high gain science return an important goal of our mission, hence by reaching a total science return of 79, which is over 51, our group had a highly successful mission.

Challenges Faced

During testing, we encountered a rocket engine failure, requiring \$20 million in repairs. Our team responded by diagnosing the issue and replacing components, ultimately improving the reliability of our launch system.

Discussion

- Test and integrate systems that utilize in-situ resources, such as lunar soil for construction or energy storage, reducing the need to transport materials from Earth.
- The greenhouse is a bioregenerative life support system, recycling air, water, and waste while growing fresh produce using hydroponics and LED lighting. Delivering the delicious and nutritious food.
- The design concept includes a cultivation area exceeding 650 m². So, it can supply 4,100 kg of edible dry mass for the mission. This study proves the substantial food production to assist astronauts on extended trips.

Mission Destination and Schedule

Our team is travelling to the Moon, for a total of 75 days, including landing and the trip back to Earth. Our justification is to have habitat on the Moon.

- November 2025: Green Run Testing
- November 2026: Hot Fire Test
- December 2026- May 2027: Static Fire Test Failure occurred post Hot-Fire Test Investigation of Failure
- June 2027-August 2027: Retesting
- March 2029: Official Launch Date
- 60 days On The Moon: Duration Of The Stay
- Late May 2029 to beginning of June 2029: Return Date

Communications



Computer Systems



Mobility



Entry, Descent, & Landing (EDL)



Mechanical Systems



Power Supply



Rockets



Science Instruments



Budget

Starting Budget:	\$100,000,000
Total Rewards:	\$251,000,000
Total Bonuses:	\$1,000,000
Unexpected Expenses:	-\$20,000,000
Total Budget:	\$351,000,000

Day 1 Purchase Total: \$0

Day 2 Purchase Total: -\$70.00

Day 3 Purchase Total: -\$95.00

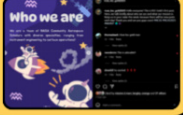
Day 4 Purchase Total: -\$103.00

Total Budget Used: -\$288.00

Lettover Budget: \$63,000,000

Outreach

Our outreach strategy includes promoting our mission and team through social media engagement, as well as pursuing collaborative opportunities with industry and academic partners.

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Budget

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Total Bonuses: \$1,000,000

Unexpected Expenses: ~\$20,000,000

Total Budget: \$351,000,000

Day 1	Day 2	Day 3	Day 4	Total
Purchase	Purchase	Purchase	Purchase	Purchase
Total: \$0	Total: -\$70.00	Total: -\$95.00	Total: -\$103.00	Total: -\$268.00
0.000	0.000	0.000	0.000	0.000

Leftover Budget: \$63,000,000

Mission Design

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The Mission Patch



Who we are



Communications



Computer Systems



Mobility



Entry, Descent, & Landing (EDL)



Mechanical Systems



Power Supply



Rockets



Science Instruments



Wild Card



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